

Tutorial class #3

Exercise 1: Acoustic radiation of a baffled piston

The purpose of this example is to study the sound field radiated by planar sources, the latter being specialized to finite radiators surrounded by effectively infinite, motionless baffles. This setting simulates for example a loudspeaker mounted on a large wall, or in the open on the ground. The planar radiator is characterized by an acceleration distribution which has specified finite values over the radiating surface; it may be either uniform (a rigid piston) or variable (structural eigenmodes of a plate *in vacuo* for instance). Here we consider the case of a rigid circular piston of radius a and uniform acceleration distribution $\gamma_0 e^{i\omega t} = i\omega V_0 e^{i\omega t}$ (see Fig. 1).

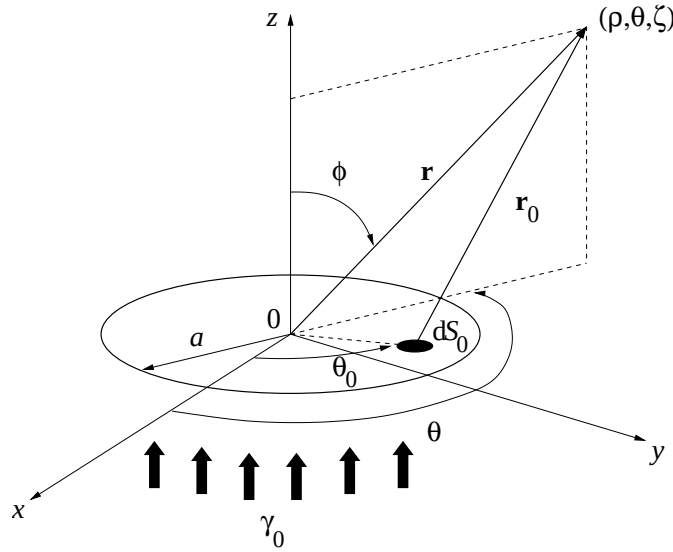


Figure 1: A circular baffled piston.

- Q1.** What is the acoustic pressure $dp(\mathbf{r}, t)$ radiated at a field point $\mathbf{r} = (\rho, \theta, \zeta)$ by an elementary surface dS_0 of the piston centered at the point $\mathbf{p}_0 = (\rho_0, \theta_0, 0)$ (in cylindrical coordinates)?
- Q2.** What is the total acoustic pressure $p(\mathbf{r}, t)$ radiated at a field point $\mathbf{r} = (\rho, \theta, \zeta)$ by the piston?
- Q3.** Compute the radiated pressure $p(\zeta, t)$ at any point $\mathbf{r} = (0, \theta, \zeta)$ on the z -axis (axis of the piston).
- Q4.** Define the far-field condition for this radiator and show that the far-field radiated pressure reads:

$$p(\mathbf{r}, t) = \frac{\rho \gamma_0 S_0}{\pi |\mathbf{r}|} e^{i(\omega t - k|\mathbf{r}|)} \frac{J_1(ka \sin \phi)}{ka \sin \phi}$$

in terms of the first-order Bessel function $z \mapsto J_1(z)$ and $S_0 = \pi a^2$, $\rho = |\mathbf{r}| \sin \phi$.

- Q5.** Define the low- and high-frequency limits in terms of ka and discuss the relevance of the previous result with that of question 3.
- Q6.** To go further (optional) *Explain how to compute the radiation impedance of the piston.*

Exercise 2: Acoustic cavity coupled to a plate¹

We consider here a coupled system constituted by a thin elastic plate clamped to a parallelepipedic acoustic cavity of which five walls apart from the plate boundary are rigid (see Fig. 2). The acoustic cavity is entirely filled with air or water, in which case no free surface exists. The fluid density is ρ_2 and the speed of sound is c_2 . The "plate" sub-system is denoted by $\Omega_1 = \{\mathbf{x} = (x, y) \in]0, a[\times]0, b[\}$, where $a > 0$ and $b > 0$ are the two dimensions of the plate of which area is thus $S_1 = ab$; its thickness is h . The material constituting the plate has density ρ_1 per unit area, Young's modulus E_1 , and Poisson's coefficient ν_1 . Concurrently, the "fluid" sub-system is denoted by $\Omega_2 = \{\mathbf{r} = (\mathbf{x}, z) \in \Omega_1 \times]0, L[\}$ and has a volume $V_2 = abL$, such that the mid-surface of the plate is located at $z = 0$. The latter is loaded by a mechanical point force of (harmonic) amplitude $F e^{-i\omega t}$ at the location $\mathbf{x}_0$ along the vertical direction $\mathbf{e}_z$.

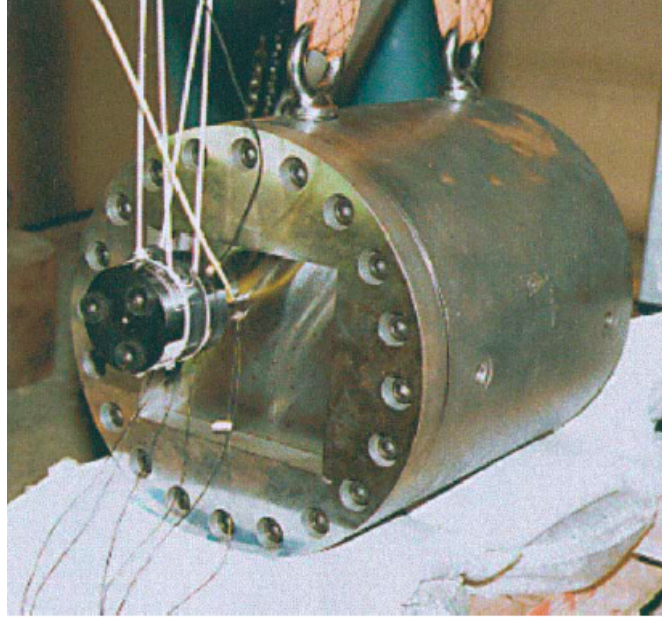


Figure 2: A parallelepipedic acoustic cavity with rigid walls coupled to a clamped thin elastic plate.

- Q1.** Write all equations satisfied by the plate bending motion $w_1(\mathbf{x}, \omega)$ in frequency domain.
- Q2.** Compute the eigenmodes of the plate *in vacuo*.
- Q3.** Derive the projection of the plate bending motion on the plate eigenmodes.
- Q4.** Write all equations satisfied by the fluid pressure $p_2(\mathbf{r}, \omega)$ in frequency domain.
- Q5.** Compute the eigenmodes of the acoustic cavity coupled to the clamped plate.
- Q6.** Derive the equations satisfied by the generalized coordinates of the fluid pressure projected on the cavity eigenmodes.
- Q7.** Derive the projection of the plate bending motion on the *cavity* eigenmodes.
- Q8.** Compute the mechanical energy of the plate.
- Q9.** Compute the acoustical energy of the cavity.
- Q10.** Compute the input power.

¹After J.-M. David, M. Menelle: Validation of a medium-frequency computational method for the coupling between a plate and a water-filled cavity, *Journal of Sound and Vibration* **265**(4), 841-861 (2003).

Important formulas

Linearized continuity equation and balance of momentum in a fluid at rest:

$$\begin{aligned}\partial_t \varrho + \operatorname{div}_{\mathbf{x}}(\varrho_{\text{ref}} \mathbf{v}) &= 0 \\ \varrho_{\text{ref}} \partial_t \mathbf{v} &= -\mathbf{grad}_{\mathbf{x}} p + \mathbf{f}.\end{aligned}$$

The acoustic pressure radiated by a pulsating single pole with a volumic flow Q (m^3/s) in a fluid of density ϱ and sound speed c :

$$p(\mathbf{x}, t) = \frac{\mathrm{i}\omega\varrho Q}{4\pi|\mathbf{x}|} e^{\mathrm{i}(\omega t - k|\mathbf{x}|)}, \quad k = \frac{\omega}{c}. \quad (1)$$

Some identities for Bessel functions $z \mapsto J_n(z)$, $z \in \mathbb{C}$, $n \in \mathbb{N}$:

$$\begin{aligned}J_n(z) &= \frac{1}{\pi} \int_0^\pi \cos(z \sin \theta - n\theta) \mathrm{d}\theta = \frac{1}{\mathrm{i}^n \pi} \int_0^\pi e^{\mathrm{i}z \cos \theta} \cos(n\theta) \mathrm{d}\theta, \\ \int_0^a J_0(kr) r \mathrm{d}r &= \frac{a J_1(ka)}{k}, \\ \frac{J_1(z)}{z} &\simeq \frac{1}{2}, \quad 1 \gg z \in \mathbb{R}.\end{aligned}$$

Solutions

Exercise 1

1. $dp(\mathbf{r}, t) = \frac{\varrho\gamma_0}{2\pi r_0} e^{i(\omega t - kr_0)} dS_0$ (twice that of the single pole because of the baffling) where $r_0 = |\mathbf{r} - \mathbf{p}_0| = |\mathbf{r}_0|$ and using $i\omega Q \equiv \gamma_0 dS_0$ in the usual formula (1).
2. $\mathbf{r}_0 = \mathbf{r} - \mathbf{p}_0$ thus $r_0^2 = r^2 + \rho_0^2 - 2\rho\rho_0 \cos(\theta - \theta_0)$, $r^2 = \rho^2 + \zeta^2$, and by Rayleigh's formula:

$$\begin{aligned} p(\mathbf{r}, t) &= \varrho\gamma_0 \int_{S_0} \frac{e^{i(\omega t - kr_0)}}{2\pi r_0} dS_0 \\ &= \varrho\gamma_0 e^{i\omega t} \int_0^a \int_0^{2\pi} \frac{e^{-ik\sqrt{r^2 + \rho_0^2 - 2\rho\rho_0 \cos(\theta - \theta_0)}}}{2\pi\sqrt{r^2 + \rho_0^2 - 2\rho\rho_0 \cos(\theta - \theta_0)}} \rho_0 d\rho_0 d\theta_0. \end{aligned}$$

3. On the piston axis $\rho = 0$, $r = \zeta$, and $p(\rho, \theta, \zeta, t) = p(\zeta, t)$ independently of θ , with:

$$\begin{aligned} p(\zeta, t) &= \varrho\gamma_0 e^{i\omega t} \int_0^a \int_0^{2\pi} \frac{e^{-ik\sqrt{\zeta^2 + \rho_0^2}}}{2\pi\sqrt{\zeta^2 + \rho_0^2}} \rho_0 d\rho_0 d\theta_0 \\ &= \varrho\gamma_0 e^{i\omega t} \int_{\zeta}^{\sqrt{\zeta^2 + a^2}} e^{-iks} ds \quad (s = \sqrt{\zeta^2 + \rho_0^2}) \\ &= \varrho c V_0 \left[e^{i(\omega t - k\zeta)} - e^{i(\omega t - k\sqrt{a^2 + \zeta^2})} \right]. \end{aligned}$$

The radiated pressure is the sum of two contributions: one coming from the center, and one coming from the boundary. This general result has a great importance in structural acoustics. The modulus of the bracketed term is displayed on Fig. 3 below as a function of $k\zeta$ for different values of ka . The axial impedance $\mathcal{Z}$ of the

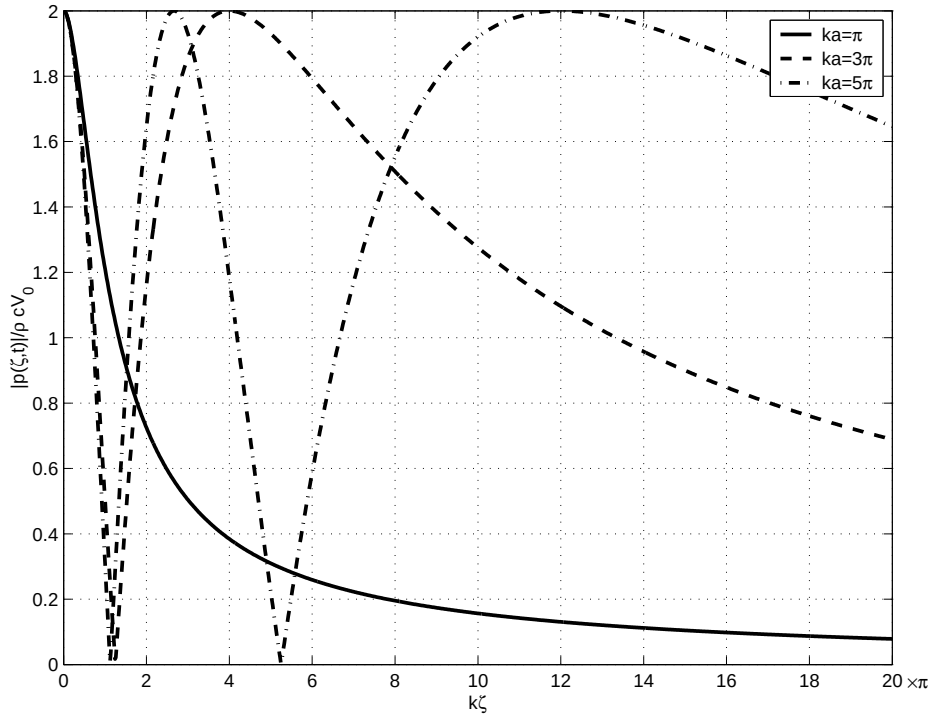


Figure 3: Radiated pressure by the piston in its axis.

piston such that $\mathcal{Z}V_0 = e^{-i\omega t} p(\zeta = 0, t)$ is:

$$\mathcal{Z}(\omega) = \varrho c (1 - e^{-ika}), \quad k = \frac{\omega}{c}.$$

4. Far-field condition $r \gg a$, then $r_0 = r - \rho_0 \sin \phi \cos(\theta - \theta_0) + O(\frac{\rho_0^2}{r^2})$ with $\sin \phi = \frac{\rho}{r}$ and:

$$\begin{aligned} p(\mathbf{r}, t) &\simeq \frac{\varrho \gamma_0}{2\pi r} e^{i(\omega t - kr)} \int_0^a \int_0^{2\pi} e^{ik\rho_0 \sin \phi \cos(\theta - \theta_0)} \rho_0 d\rho_0 d\theta_0 \\ &= \frac{\varrho \gamma_0}{r} e^{i(\omega t - kr)} \int_0^a J_0(k\rho_0 \sin \phi) \rho_0 d\rho_0 \\ &= \frac{\varrho \gamma_0}{r} e^{i(\omega t - kr)} \frac{a J_1(ka \sin \phi)}{k \sin \phi} \end{aligned}$$

using the identities for Bessel functions recalled with the "Important formulas." In terms of the single pole radiated pressure ($i\omega Q \equiv \pi a^2 \gamma_0 = S_0 \gamma_0$) this is:

$$p(\mathbf{r}, t) \simeq 4p_s(\mathbf{r}, t) \frac{J_1(ka \sin \phi)}{ka \sin \phi}.$$

The function $z \mapsto z^{-1} J_1(z)$ (called directivity factor) is plotted on Fig. 4 below for $z \equiv ka \sin \phi \in \mathbb{R}_+$. Near the piston axis $\phi \simeq 0$ or for $ka \ll 1$ (low-frequency limit) one then has $p(\mathbf{r}, t) \simeq 2p_s(\mathbf{r}, t)$ and $|p(\mathbf{r}, t)| \simeq \frac{\varrho S_0 \gamma_0}{2\pi |\mathbf{r}|}$. The directivity function $\phi \mapsto D(\phi)$ with:

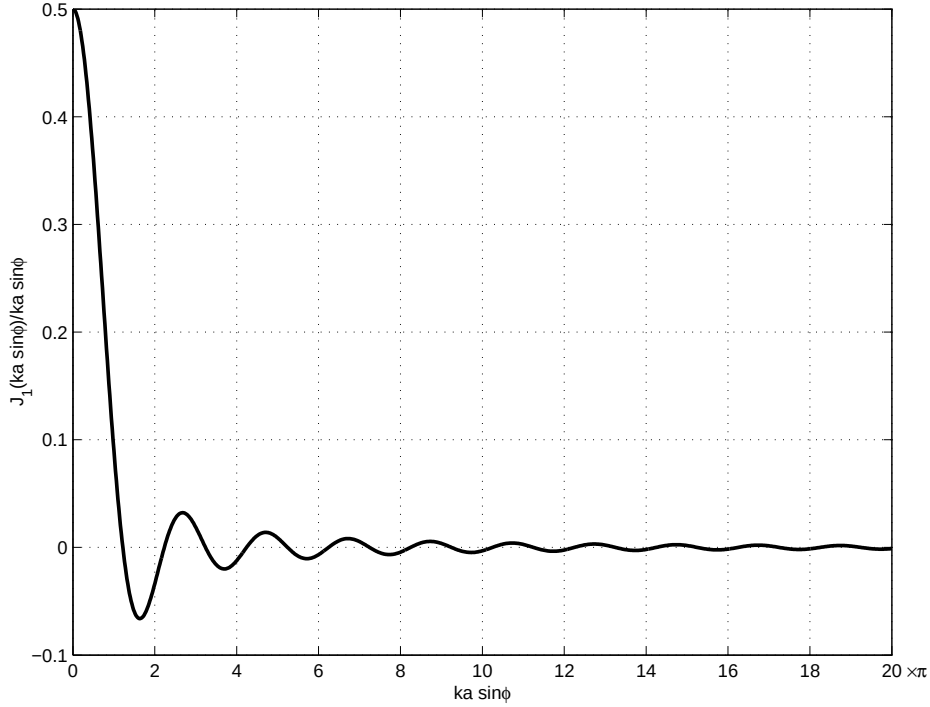


Figure 4: $z \mapsto \frac{J_1(z)}{z}$ for $z \equiv ka \sin \phi$.

$$D(\phi) = \frac{2J_1(ka \sin \phi)}{ka \sin \phi}$$

is displayed in Fig. 5 for several values of the frequency parameter ka .

5. Low-frequency range $ka \ll 1$, high-frequency range $ka \gg 1$. The modulus of the far-field ($a \ll \zeta$) radiated pressure on the piston axis reads:

$$\begin{aligned} |p(\zeta, t)| &= 2\varrho c \left| V_0 \sin \frac{k}{2} \left(\sqrt{a^2 + \zeta^2} - \zeta \right) \right| \\ &= 2\varrho c \left| V_0 \sin ka \left[\frac{a}{4\zeta} + O\left(\frac{a^3}{\zeta^3}\right) \right] \right| \\ &\simeq \frac{\varrho S_0 \gamma_0}{2\pi \zeta}. \end{aligned}$$

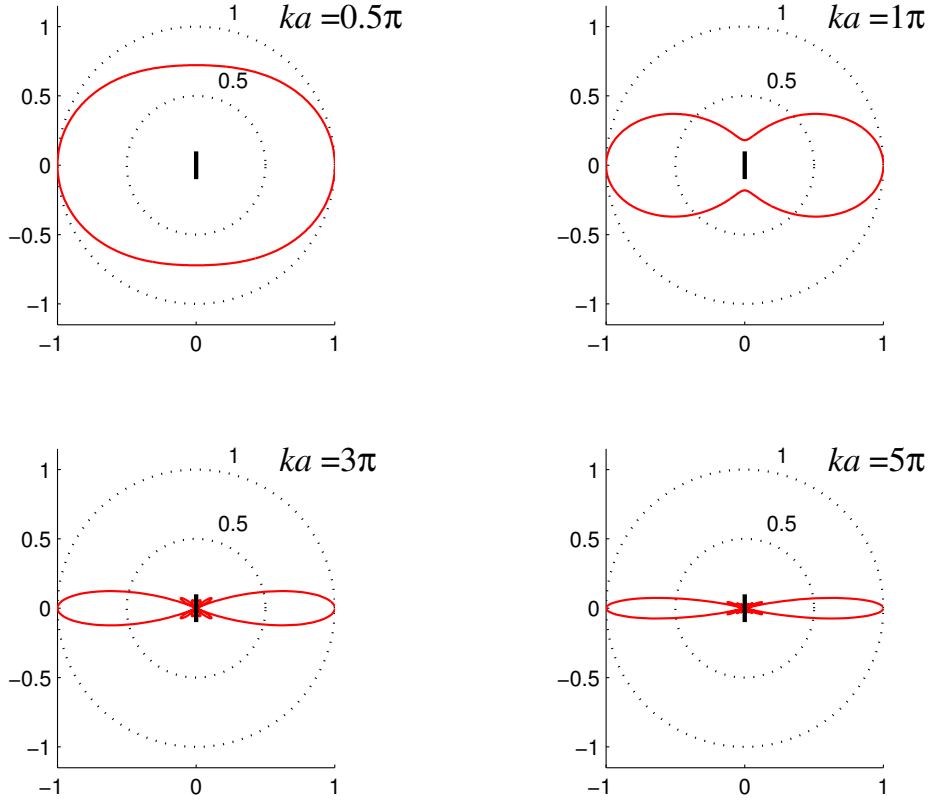


Figure 5: The directivity function $\phi \mapsto D(\phi)$ for different frequency parameters ka .

This is coherent with the result of question 4 with $\phi = 0$.

6. The impedance $\mathcal{Z}(\omega)$ is the ratio of the overall force exerted by the fluid on the piston, and the speed of the piston:

$$\begin{aligned}
 i\omega \mathcal{Z}(\omega) &= \frac{i\omega}{V_0 e^{i\omega t}} \int_0^a \int_0^{2\pi} p(\rho, \theta, 0, t) \rho d\rho d\theta \\
 &= -\varrho \omega^2 \int_0^a \int_0^{2\pi} \int_0^a \int_0^{2\pi} \frac{\rho \rho_0 e^{-ik\sqrt{\rho^2 + \rho_0^2 - 2\rho\rho_0 \cos(\theta - \theta_0)}}}{2\pi \sqrt{\rho^2 + \rho_0^2 - 2\rho\rho_0 \cos(\theta - \theta_0)}} d\rho_0 d\theta_0 d\rho d\theta \\
 &\equiv -\omega^2 M_p\left(\frac{\omega}{c}\right) + i\omega D_p\left(\frac{\omega}{c}\right)
 \end{aligned}$$

from the result of question 2. The reactive part $M_p\left(\frac{\omega}{c}\right) = \Re \frac{\mathcal{Z}(\omega)}{i\omega}$ corresponds to an added mass (of *a priori* unknown sign), and the resistive part $D_p\left(\frac{\omega}{c}\right) = \Re \mathcal{Z}(\omega)$ (which is always positive) corresponds to an added damping effect, in fact the radiation of some part of the energy of the piston into the acoustic medium—this is not a true dissipation. They may be computed explicitly as (see M.C. Junger & D. Feit, *Sound, Structures, and their Interaction*, ASA 1993, pp. 106-109):

$$M_p(k) = \varrho a S_0 \frac{H_1(2ka)}{(ka)^2}, \quad D_p(k) = \varrho c S_0 \left[1 - \frac{J_1(2ka)}{ka} \right],$$

where:

$$z \mapsto H_n(z) = \frac{z^n}{2^{n-1} \sqrt{\pi} \Gamma(n + \frac{1}{2})} \int_0^1 (1 - t^2)^{n-\frac{1}{2}} \sin(zt) dt$$

is the Struve function of order n . The normalized added mass function $ka \mapsto \frac{M_p(k)}{\varrho a S_0}$ and added damping function $ka \mapsto \frac{D_p(k)}{\varrho c S_0}$ (called *radiation efficiency* in the structural-acoustics literature) are plotted on Fig. 6 below.

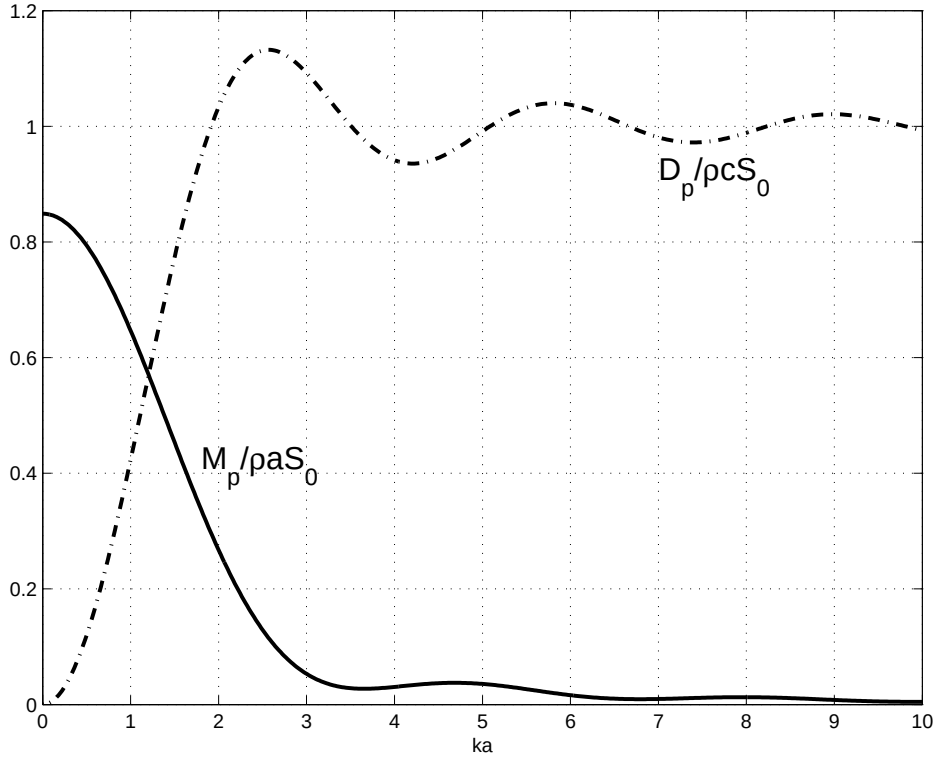


Figure 6: Normalized reactive and resistive parts of the radiation impedance of a circular piston.

Exercise 2

1. The thin plate bending equation in frequency domain reads (Kirchhoff-Love kinematics):

$$-\rho_1 \omega^2 w_1 + D_1 \Delta_x^2 w_1 = F \delta(\mathbf{x} - \mathbf{x}_0) - p(\mathbf{x}, \omega),$$

where $p(\mathbf{x}, \omega) = p_2(\mathbf{x}, z = 0, \omega)$, and $D_1 = \frac{E_1 h^3}{12(1-\nu_1^2)}$ where E_1 is Young's modulus of the plate material, ν_1 is its Poisson's coefficient, and ρ_1 its mass per unit area. The boundary conditions are (clamped plate):

$$\begin{aligned} w_1(x=0, y, t) &= w_1(x=a, y, t) = 0, & \forall y \in]0, b[, \\ w_1(x, y=0, t) &= w_1(x, y=b, t) = 0, & \forall x \in]0, a[, \\ \partial_x w_1(x=0, y, t) &= \partial_x w_1(x=a, y, t) = 0, & \forall y \in]0, b[, \\ \partial_y w_1(x, y=0, t) &= \partial_y w_1(x, y=b, t) = 0, & \forall x \in]0, a[. \end{aligned}$$

2. The eigenmodes of the plate *in vacuo* have the form:²

$$\phi_{1,\alpha}(\mathbf{x}) = \psi_{\alpha_1}\left(\frac{x}{a}\right) \times \psi_{\alpha_2}\left(\frac{y}{b}\right),$$

$x \in [0, a]$, $y \in [0, b]$, $\alpha = (\alpha_1, \alpha_2) \in (\mathbb{N} \setminus \{0, 1\})^2$, with:

$$\psi_n(x) = A e^{i\gamma_n x} + B e^{-i\gamma_n x} + C e^{\gamma_n x} + D e^{-\gamma_n x}.$$

The wave numbers γ_n are shown to satisfy $\cosh \gamma_n \cos \gamma_n = 1$, which also reads $\tanh^2 k_n = \tanh^2 k_n$ letting $k_n = \frac{\gamma_n}{2}$. One thus ends up with (see Fig. 7):

$$\begin{aligned} \tanh k_n + \tan k_n &= 0 & \text{if } n \text{ is even,} \\ \tanh k_n - \tan k_n &= 0 & \text{if } n \text{ is odd.} \end{aligned}$$

Consequently:

²See for instance G. B. Warburton: The vibrations of rectangular plates, *Proceedings of the Institution of Mechanical Engineers* **168**(12), 371-381 (1954).

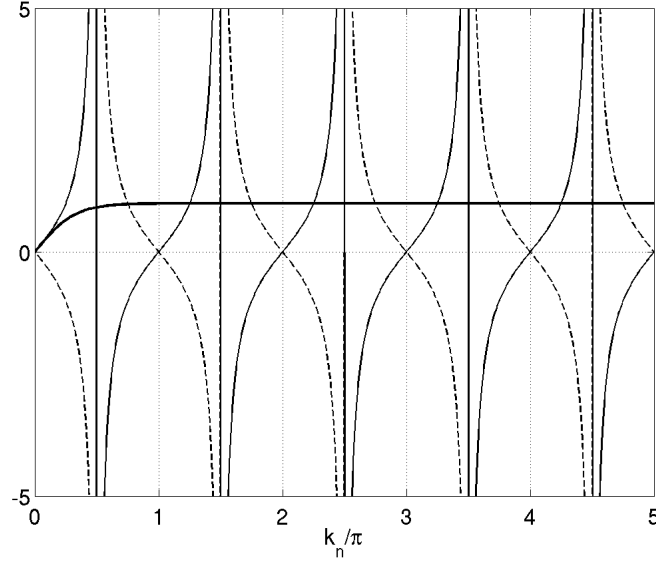


Figure 7: Plots of the functions $x \mapsto \tanh x$ (—), $x \mapsto \tan x$ (---), and $x \mapsto -\tan x$ (....).

$$\psi_n(x) = \begin{cases} \cos k_n(2x-1) + \frac{\sin k_n}{\sinh k_n} \cosh k_n(2x-1) & \text{if } n \text{ is even,} \\ \sin k_n(2x-1) - \frac{\sin k_n}{\sinh k_n} \sinh k_n(2x-1) & \text{if } n \text{ is odd,} \end{cases}$$

where the k_n 's are the positive solutions of $\tan k_n + (-1)^n \tanh k_n = 0$, $n \geq 2$. The associated eigenfrequencies are:

$$\omega_\alpha = (\lambda_\alpha \pi)^2 \sqrt{\frac{D_1}{\varrho_1}}, \quad \lambda_\alpha^4 = \frac{A_{\alpha_1}^4}{a^4} + \frac{A_{\alpha_2}^4}{b^4} + \frac{2}{a^2 b^2} [\nu_1 B_{\alpha_1} B_{\alpha_2} + (1 - \nu_1) C_{\alpha_1} C_{\alpha_2}],$$

where A_n, B_n, C_n are coefficients depending on the boundary conditions (clamped plate here).

3. Then the plate bending motion projected on the eigenmodes reads:

$$w_1(\mathbf{x}, \omega) = \sum_{|\alpha|=4}^{\infty} \frac{F \phi_{1,\alpha}(\mathbf{x}_0) - \int_{\Omega_1} \phi_{1,\alpha}(\mathbf{y}) p(\mathbf{y}, \omega) d\mathbf{y}}{\mu_\alpha (\omega_\alpha^2 - \omega^2 + 2i\xi_1 \omega_\alpha \omega)} \phi_{1,\alpha}(\mathbf{x}),$$

where ξ_1 is the critical damping rate of the plate, assumed to be identical for all modes, and $\mu_\alpha = \int_{\Omega_1} \varrho_1 \phi_{1,\alpha}^2(\mathbf{y}) d\mathbf{y}$ is the generalized mass of the α -th mode.

4. For the fluid pressure field $p_2(\mathbf{x}, z, \omega)$ one has:

$$\begin{aligned} (\Delta + k_2^2) p_2 &= 0 & \text{in } \Omega_2, \\ \partial_z p_2 &= \varrho_2 \omega^2 w_1 & \text{for } z = 0, \\ \partial_n p_2 &= 0 & \text{on rigid walls,} \end{aligned}$$

with $k_2 = \frac{\omega}{c_2}$ (the acoustic wave number).

5. The eigenmodes of the cavity are:

$$\begin{aligned} \phi_{2,\beta}(\mathbf{x}, z) &= \frac{2}{\sqrt{ab}} \cos\left(\beta_1 \pi \frac{x}{a}\right) \times \cos\left(\beta_2 \pi \frac{y}{b}\right) \times f_\beta(z) \\ &= \varphi_{2,\beta}(\mathbf{x}) \times f_\beta(z), \end{aligned}$$

$\mathbf{x} \in [0, a] \times [0, b]$, $z \in [0, L]$, $\beta = (\beta_1, \beta_2) \in \mathbb{N}^2$, with:

$$f_\beta(z) = \cosh k_z(L - z)$$

and:

$$k_z^2 = \left(\frac{\beta_1 \pi}{a} \right)^2 + \left(\frac{\beta_2 \pi}{b} \right)^2 - k_2^2.$$

6. The fluid pressure projected on the cavity eigenmodes reads:

$$p_2(\mathbf{x}, z, \omega) = \sum_{|\beta|=0}^{\infty} q_\beta(\omega) \varphi_{2,\beta}(\mathbf{x}) f_\beta(z),$$

where the generalized coordinates $\{q_\beta\}_{\beta \in \mathbb{N}^2}$ are deduced from the boundary condition:

$$\frac{\partial p_2}{\partial z} = \varrho_2 \omega^2 w_1, \quad z = 0.$$

Introducing the coupling coefficients (see Lecture #5):

$$\gamma_{\alpha\beta} = \int_{\Omega_1} \varrho_2 \phi_{1,\alpha}(\mathbf{x}) \varphi_{2,\beta}(\mathbf{x}) d\mathbf{x},$$

one obtains from the foregoing boundary condition projected on $\phi_{1,\alpha}$:

$$\sum_{|\beta|=0}^{\infty} \gamma_{\alpha\beta} \left[f_\beta(0) + \frac{\varrho_1}{\varrho_2} \left(\frac{\omega_\alpha^2}{\omega^2} - 1 + 2i\xi_1 \frac{\omega_\alpha}{\omega} \right) f'_\beta(0) \right] q_\beta(\omega) = \varrho_2 F \phi_{1,\alpha}(\mathbf{x}_0), \quad \forall |\alpha| \geq 4.$$

The coupling coefficients $\gamma_{\alpha\beta}$ can be directly computed from the foregoing expressions of the plate and cavity eigenmodes. One obtains $\gamma_{\alpha\beta} = 2\varrho_2 \sqrt{ab} \times G_{\alpha_1\beta_1} \times G_{\alpha_2\beta_2}$ where:

$$G_{np} = \begin{cases} (1 + (-1)^p) \frac{16k_n^3 \sin k_n}{16k_n^4 - (p\pi)^4} & \text{if } n \text{ is even,} \\ (1 - (-1)^p) \frac{16k_n^3 \cos k_n}{16k_n^4 - (p\pi)^4} & \text{if } n \text{ is odd.} \end{cases}$$

7. Using again the plate-fluid boundary condition yields:

$$w_1(\mathbf{x}, \omega) = \frac{1}{\varrho_2 \omega^2} \sum_{|\beta|=0}^{\infty} q_\beta(\omega) \varphi_{2,\beta}(\mathbf{x}) f'_\beta(0).$$

8. The bending energy of a thin plate (Kirchhoff-Love kinematic) is:

$$\mathcal{E}_1(\omega) = \frac{1}{2} \int_{\Omega_1} \varrho_1 \omega^2 |w_1|^2 d\mathbf{x} + \frac{1}{2} \int_{\Omega_1} \mathbb{M} : \mathbb{K} d\mathbf{x}$$

where $\mathbb{K} = -\nabla_{\mathbf{x}} \otimes \nabla_{\mathbf{x}} w_1$ is the second-order tensor of curvature variations and $\mathbb{M} = \nu_1 D_1(\text{tr} \mathbb{K}) \mathbb{I}_2 + (1 - \nu_1) D_1 \mathbb{K}$ is the second-order tensor of bending moments; $\mathbb{I}_2$ is the 2×2 identity matrix. Owing to the above expansion one has:

$$\mathcal{E}_1(\omega) = \frac{\varrho_1}{2\varrho_2^2 \omega^2} \sum_{|\beta|=0}^{\infty} \left(1 + \frac{k_\beta^4}{k_1^4(\omega)} \right) |f'_\beta(0)|^2 |q_\beta(\omega)|^2,$$

with $k_\beta^2 = \left(\frac{\beta_1 \pi}{a} \right)^2 + \left(\frac{\beta_2 \pi}{b} \right)^2$ and $k_1^2(\omega) = \omega \sqrt{\frac{\varrho_1}{D_1}}$ the flexural wave number of the plate.

9. The acoustical energy of the cavity is given by:

$$\mathcal{E}_2(\omega) = \frac{1}{2} \int_{\Omega_2} \varrho_2 |\mathbf{v}_2|^2 d\mathbf{x} dz + \frac{1}{2} \int_{\Omega_2} \frac{|p_2|^2}{\kappa_2} d\mathbf{x} dz,$$

where $\kappa_2 = \varrho_2 c_2^2$ is the compressibility of the fluid, and $\mathbf{v}_2$ is the fluid velocity field given in terms of the fluid pressure p_2 by $i\varrho_2 \omega \mathbf{v}_2 = -\nabla p_2$. One finally obtains:

$$\mathcal{E}_2(\omega) = \frac{1}{2\varrho_2 \omega^2} \sum_{|\beta|=0}^{\infty} \left(\int_0^L [(k_\beta^2 + k_2^2) |f_\beta(z)|^2 + |f'_\beta(z)|^2] dz \right) |q_\beta(\omega)|^2.$$

10. The input power is:

$$\begin{aligned}\Pi_{\text{IN}}(\omega) &= -\Re\{\mathrm{i}\omega\bar{w}_1(\boldsymbol{x}_0, \omega)F\} \\ &= -\frac{F}{\varrho_2\omega} \sum_{|\beta|=0}^{\infty} \Im\{f'_\beta(0)q_\beta(\omega)\}\varphi_{2,\beta}(\boldsymbol{x}_0).\end{aligned}$$